

Chittagong University of Engineering and Technology



Department of Electrical and Electronic Engineering

Project Report

Project Title: Design and
Implementation of Gas Leakage and
Fire Alarm System

Course Code : EEE 222

Course Title : Circuit Simulation and Shop Practice

Submitted to :

Mrinmoy Dey
Associate Professor,
Department of Electrical and Electronic Engineering,
Chittagong University of Engineering and Technology

Submitted By :

Group No : 01

Members:

- 1. Name** : Mohammed Tawshif Hossain
Student ID : 2002001
Section : A
- 2. Name** : Salman Asif
Student ID : 2002031
Section : A

Design and Implementation of Gas Leakage and Fire Alarm System

Mohammed Tawshif Hosssain
Department of Electrical and
Electronic Engineering
Chittagong University of
Engineering and Technology
Raozan, Chittagong-
4349, Bangladesh
u2002001@student.cuet.ac.bd

Salman Asif
Department of Electrical and
Electronic Engineering
Chittagong University of
Engineering and Technology
Raozan, Chittagong-
4349, Bangladesh
u2002031@student.cuet.ac.bd

Associate professor Mrinmoy Dey
Department of Electrical and
Electronic Engineering
Chittagong University of
Engineering and Technology
Raozan, Chittagong-
4349, Bangladesh
mrinmoy@cuet.ac.bd

Abstract— The significance of a home safety detection system cannot be overstated when it comes to ensuring the security of individuals. Continuously monitoring appliances such as LPG gas cylinders, electrical components, and potential short circuits in both residential and large-scale industrial settings can be challenging. Additionally, in densely populated areas like markets, bazaars, and shopping malls, immediate detection of gas leakage and fire hazards is crucial to prevent catastrophic incidents. The "Gas Leakage and Fire Alarm System" described here is specifically designed to detect gas leaks and fires, thus mitigating the risk of explosions and maintaining overall safety. The system incorporates an MQ5 gas sensor for detecting gas leaks and fires in the surrounding environment. An Arduino unit is used to display gas and fire levels, providing clear visual information. Moreover, the system is equipped with audible alerts, ensuring timely warnings upon detecting a gas leakage or fire, thus enhancing overall safety measures.

Keywords: *Arduino Uno, MQ5 Gas sensor, Embedded system, Buzzer.*

I. Introduction

In this rapidly evolving society, electronics has firmly established itself as the paramount field of Engineering. Electronic devices are extensively employed across various industries to ensure quality control and automation, gradually substituting the current extensive workforce involved in processing and assembly within factories. These devices augment human capabilities to perceive, monitor, compute, observe and regulate. [8]

Natural gas is an odorless one which consists of compounds made of two elements: carbon and hydrogen called hydrocarbons. The unregulated fusion of fuel and oxygen unleashes thermal energy, and any unintentional interaction with a spark can initiate ignition. Automated gas sensing systems are intended to inform the inhabitants of a building

or factory to vacate during instances of gas leakage or other emergencies. Furthermore, these systems aim to relay the incident to an external location in order to request the presence of emergency services, as well as to ready the structure and related systems to mitigate the propagation of gas and fire. [3]

This project focuses on the development and building of a gas detection system equipped with an audible system, arduino and gas sensor. The primary objective of this gas detector, accompanied by an audible alarm, is to enhance the intelligent detection, timely notification and interactive management of gas incidents, surpassing the capabilities of traditional fire alarm system.

There are two prevalent variants of smoke detectors: ionization and photoelectric. Caution must be exercised when choosing the appropriate detector type. Ion detectors are more adept at detecting flaming wires, while photoelectric detectors tend to identify smoldering fires more rapidly in the majority of situations. In buildings, manual fire alarm boxes are typically positioned at every exit on each floor, at the very least. In the presence of an automatic sprinkler system, waterflow devices are utilized to indicate the functioning of the system. [1]

Several researches are conducted on making suitable, effective gas leakage system. Different approaches like embedded system, infrared principle are used to design the system. In order to address the challenges posed by gas accidents in coal mines and to enhance family safety in gas usage, a novel infrared detection system with integration and miniaturization has been developed. The underlying principle of infrared detection optics used in this system is extensively examined. The concept of multi-gas detection is introduced and guided by analyzing single gas detection. By

researching the design of the cell structure, a compact and integrated cell has been devised. The data transmission method via the Controller Area Network (CAN) bus is elucidated. The gas detection system's functional block is designed with its hardware and software systems analyzed and devised, utilizing a single-chip microcontroller for intelligent processing. This designed system meets the technological requirements of low power consumption, compact size, wide measurement range and the capability to detect multiple gases. [15]

Another research focuses on the environmental consideration. The growing demand for environmental pollution monitoring and control systems has led to the development of an embedded system using PIC 16F877 Microcontroller for the detection of hazardous gas leaks. This system offers portability, affordability and ease of implementation, making it suitable for both chemical industries and residential areas near such industries. By monitoring gases like LPG and propane, the system can rapidly detect abnormal gas levels and trigger an alarm while sending an alert message via SMS to authorized users. Real time monitoring of gas concentration allows for proactive measures to avoid endangering human lives. Overall, the system provides an efficient solution for detecting and addressing hazardous gas leaks, ensuring the safety of individuals and preventing potential harm. [10]

II. Literature review:

An automatic gas alarm system is designed to detect fire by monitoring environmental changes related to combustion. Fire alarm systems can be automatically or manually actuated, or both. Automatic systems notify occupants to evacuate, alert off-site locations for emergency services, and prepare the building to control the fire's spread. [5]

The primary power supply for a fire alarm system is typically a non-switched 120 or 240 Volt Alternating Current obtained from a commercial power utility. In non-residential settings, a separate branch circuit is dedicated solely to the fire alarm system and its components. It's important to note that "dedicated branch circuits" should not be confused with "individual branch circuits" which supply power to a single appliance. To ensure uninterrupted operation, secondary power supplies such as sealed lead-acid storage batteries or emergency sources like generators are utilized to provide backup power in case of a primary power failure.

A microcontroller is a self-contained system that includes a processor, memory, and peripherals. It can be used in conjunction with an embedded system, where only the software needs to be added. Many microcontrollers are embedded within various machinery such as automobiles, telephones, appliances, and computer peripherals, forming what are known as embedded systems. Embedded systems can range from sophisticated setups to simpler ones with minimal memory and program requirements, lacking an operating system and having low software complexity. Common input and output devices in embedded systems

include switches, relays, solenoids, LEDs, LCD displays, radio frequency devices, and sensors for data collection (e.g., temperature, humidity, light level). [6] Unlike personal computers, embedded systems typically lack recognizable I/O devices like keyboards, screens, disks, printers, and may not include human interaction devices. The choice of detectors and fire alarm system for property protection depends on the owner's goals, property value, and insurance company requirements. [1]

[12] In 2014, it was emphasized that a reliable controller is crucial for continuously detecting gas leakage and monitoring gas levels. A fast response is necessary when a leakage is detected, and the monitoring system should provide additional information about the leakage for further processing. The detection system mentioned in the study involved an Arduino, a microcontroller board compatible with ATmega328p. The system was coupled with a weight sensor, an LCD display, GSM capabilities, and a DC motor. These components worked together to ensure efficient and accurate gas detection and monitoring.

In their work, [4] developed and implemented a GSM-based remote monitoring system. The focus of the study was on wireless monitoring, as it offers broader applications. The system's hardware and software architecture were designed to transmit remote signals through the GSM network. The system comprised two parts: the monitoring center and the remote monitoring station. The monitoring center included a computer and a TC35 communication module for GSM, connected via RS232. The remote monitoring station consisted of a TC35 communication module for GSM, an MSP430F149 MCU, a display unit, sensors, and a data gathering and processing unit. The software for both the monitoring center and the remote monitoring station was developed using Visual Basic (VB). This GSM-based remote monitoring system provided a means for remote monitoring and data processing.

Finally several methods can be used to build the product. But what important here is the marketing and implementation of the product to the consumer level. The product usage module, how to set up the product, cost effect all should be in the consideration.

III. Methodology

The schematic diagrams and block diagram depict the whole gas leakage system from power stage to detection. The amount of supplied power into the circuit is dependent on the combined power usage on each component in the circuit. From the diagram, we can see that the negative polarity of 9V battery is connected to the ground of MQ5 Gas sensor, Ground of Arduino, Vss and K pin of LCD Display. Then the positive polarity is connected to the Vin pin of Arduino. The Vcc of Gas sensor is connected to 5v pin of Arduino, VDD and a resistor(4.7k) which is connected to the V0 pin of the LCD. Then another resistor of 1k is short with the 4.7k resistor and ground of the gas sensor. The 100 ohm resistor is connected with the VCC of Gas sensor and A pin of the LCD Display. It is also short with the 4.7K resistor.

Then the buzzer is connected with the 13 and Ground pin of the Arduino. Two LED are used and they are connected along with the two 100ohm resistor with Arduino and LCD display.

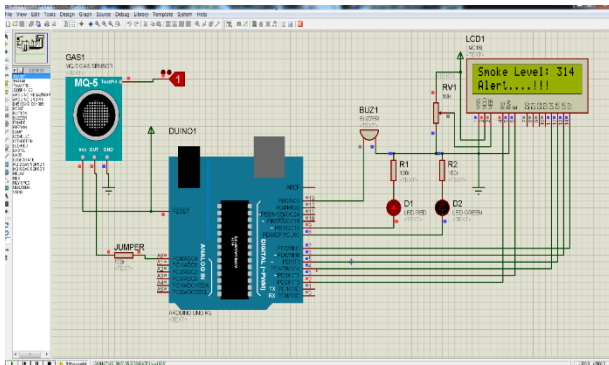


Figure 01: Proteus circuit diagram of gas leakage and fire alarm system.

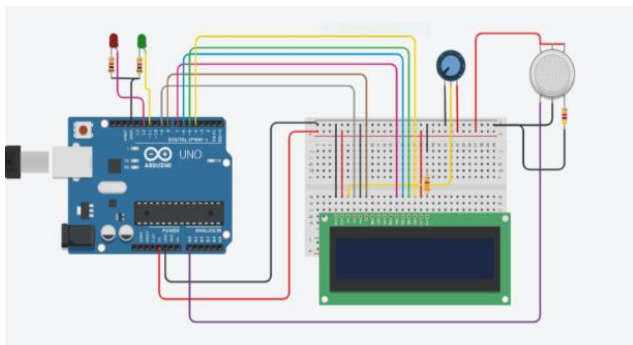


Figure 02: Complete circuit diagram of the following project.

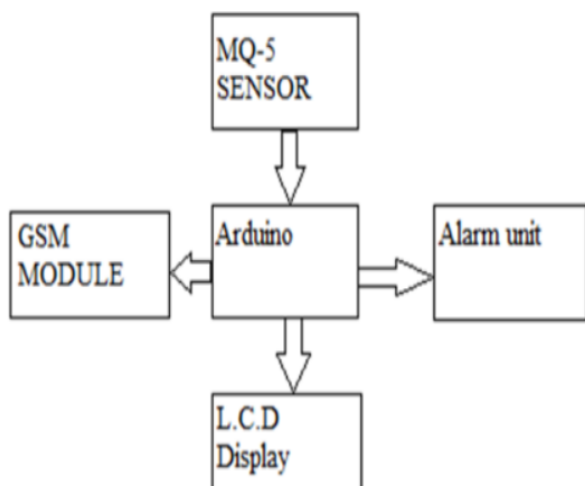


Figure 03: Block diagram of the Gas Leakage system.

The Blue LED is the indication of power supply. If we connect the 9v battery to the system then the Blue LED

will be on. Then comes the Gas sensor. It is built around a timer to accept input from the Gas sensor MQ5 and activate a Buzzer and set of Red LED and ALERT in the LCD display via Arduino during the gas leakage.

Hardware Requirement:

Here are following requirements for the project:

- Arduino Uno
- MQ5 Gas Sensor
- LCD Display
- Breadboard
- Resistor(100):3
- Resistor (4.7k): 1
- Resistor (1k);1
- LED Light(Red and Blue)
- Jumper wire (Male to Female)

Arduino Uno:

The Arduino Uno is a popular open-source microcontroller board developed by Arduino.cc. It was first released in 2010 and is based on the Microchip ATmega328P microcontroller. The board features a combination of digital and analog input/output (I/O) pins, which can be connected to various expansion boards and circuits. It offers 14 digital I/O pins, six of which support PWM output, and six analog I/O pins. The board can be programmed using the Arduino IDE and is connected to a computer via a type B USB cable. Power can be supplied through either a USB cable or a barrel connector, accepting voltages between 7 and 20 volts. The Arduino Uno shares the same microcontroller as the Arduino Nano board and has similar headers to the Leonardo board. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license, and the Arduino website provides access to layout and production files for some hardware versions.

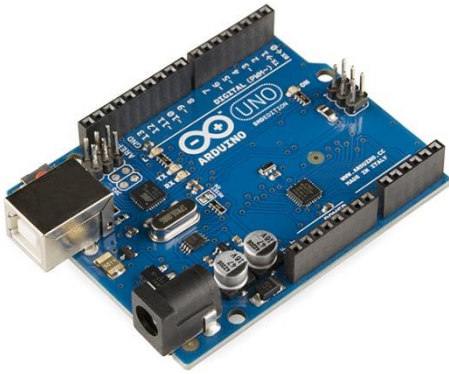


Figure 04: Arduino Uno

MQ-5 Gas sensor:

This MQ-5 Methane LPG Liquid Propane Gas Sensor module is widely used in gas leakage detecting pieces of equipment in family and industry, are suitable for detecting LPG, natural gas, town gas, avoid the noise of alcohol and cooking fumes and cigarette smoke. The MQ5 (MQ-5) is used in gas leakage detecting equipment in consumer and industry applications, this sensor is suitable for detecting LPG, natural gas, coal gas. Avoid the noise of alcohol, cooking fumes, and cigarette smoke. The sensitivity can be adjusted by the potentiometer. The sensitive material of the MQ-5 gas sensor is tin Oxide (SnO_2), which with lower conductivity in clean air. When the target combustible gas exists, the sensor's conductivity is higher along with the gas concentration rising. It will take analog output from the arduino. When there is gas or fire the voltage will be high otherwise the voltage will be low.



Figure 05: MQ-5 Gas sensor

Breadboard:

Breadboard has sets of miniature sockets laid out on a 0.1 inch grid which will accept the manual insertion of component leads and tinned copper wire (TCW) links (Figure 10.6a). It has rows of contacts interconnected in groups placed either side of the center line of the board, where the integrated circuits (ICs) are inserted, giving multiple contacts on each IC pin. At each side of the board, there are long rows of common contacts, which are used for the power supplies. Some types of breadboard are supplied in blocks that link together to accommodate larger circuits, or are mounted on a base with built-in power supplies. This the construction base for the building of electronic circuits.

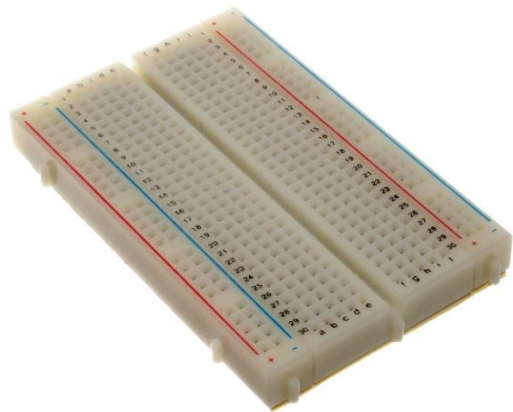


Figure 06: Breadboard

LCD Display:

An LCD (Liquid Crystal Display) display is a type of flat-panel display commonly used in electronic devices such as calculators, digital watches, televisions, computer monitors, and various other devices. LCD displays are preferred for their low power consumption, thin profile, and ability to display content with high contrast and sharpness. LCD displays consist of multiple layers, including a backlight source, a polarizing filter, a layer of liquid crystal material, and a second polarizing filter.



Figure 07: LCD Display

Resistors:

The 100, 4.7k, 1k resistors are used as passive electronic component used to control the flow of electric current in the circuit. They are used to limit the current flow in a circuit to a specific level.

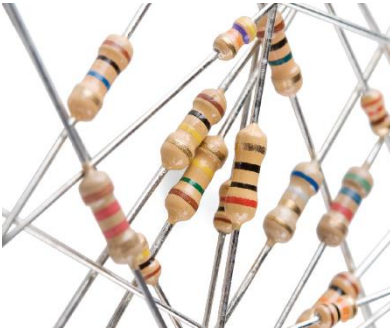


Figure 08: Resistors

LED:

LED stands for Light Emitting Diode. It is a semiconductor device that emits light when an electric current passes through it in the forward direction. LEDs are commonly used in various applications, including lighting, electronic displays, indicators, and automotive lighting.

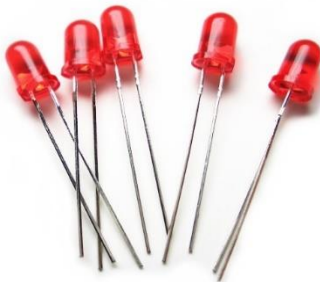


Figure 09: LED

Battery:

A 9V battery is used in the system to power the whole circuit. It is connected to Vin port of Arduino. All the other electronic components absorb energy from the 9V battery.



Figure 10: Battery (9V)

The overall project prototype is given below:

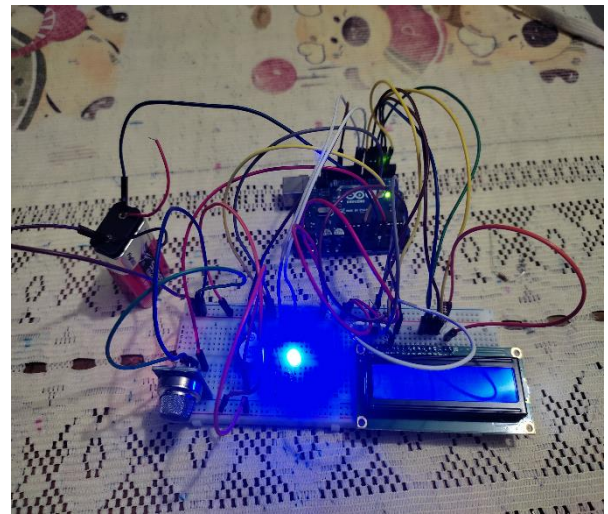


Figure 11: Final project prototype of Gas Leakage and Fire Alarm system

IV. Result

The final product is ready to work. If we connect the 9V battery to the system the whole system will be on and the indication is the blue LED. If there is smoke or gas in the atmosphere within the range, the gas sensor will detect the gas and the alarm or buzzer will be on.

V. Cost analysis

The total cost of our project is given below with data sheet.

Component	Description	Unit cost	Total cost
Arduino Uno	Atmega 328p Microcontroller to take input and give output	1000	1000
MQ-5 Gas Sensor	A sensor of detecting LPG Gas and fire	150	150
LCD Display	To display the output	200	200
Buzzer	Audible system to detect the fire and notify	15	15
Jumper (Male to Male)	Connect all the component in the breadboard to flow current	3	60
Breadboard	Arranging the whole system	120	120
Battery (9V)	Power the system	70	70
Resistors	To control current flow to the components	5	15

VI. Applications

Based on the findings of this research, several important implications can be drawn. Firstly, this design holds significant potential in environments where individuals are either hearing-impaired or speech-impaired, as the combination of vibrant red LED lights and prominent danger signs serves as a highly effective means of signaling imminent hazards. Secondly, for occupants who are visually impaired, the inclusion of a buzzer alert system provides a valuable method for alerting them to potential dangers. This innovative solution offers an aesthetically pleasing and practical approach to address the needs of diverse populations, ensuring safety and effective communication in various scenarios.

VII. Limitations

The system cannot be controlled from the outside. That means to control someone must be there with the system. Also sometime the gas sensor cannot detect the gas level perfectly. So strong gas sensor must be used.

VIII. Future addition

Here we can use temperature sensor to measure over temperature and can give precaution in the industry. We can implement automatic gas off features and can create a wireless model of it and can connect it with the mobile app so that the occupants from the outside can get notification. In the big market or industry immediate fire extinguisher can cause reduction of fatal incidents. So we can integrate fire extinguisher with the system and design the code to implement the plan.

IX. Acknowledgement

We would like to express our sincere gratitude and appreciation to Associate Professor Mrinmoy Dey sir for his invaluable guidance, expertise, and unwavering support throughout the course of this academic project. The extensive knowledge, insightful suggestions, and constructive feedback have played an instrumental role in shaping and refining this project. We are truly grateful for the mentorship and the time dedicated to reviewing and providing valuable inputs. Furthermore, we extend our heartfelt thanks to Chittagong University of Engineering and Technology for providing us with the necessary resources, facilities, and conducive academic en

X. Conclusion

The implemented gas detector system exhibited several favorable characteristics, including affordability, wide applicability, design efficiency, availability of locally sourced components and research materials, compatibility, portability, and durability. The project's performance during testing met the specified design requirements. However, it is important to note that the overall operation and performance of the system rely on the user, who may be susceptible to human errors such as entering incorrect timing. The construction of the system was carefully executed to facilitate easy maintenance and repairs, ensuring that any potential breakdown can be addressed in an affordable manner for the user.

XI. References

- [1] A Practical Guide to Fire Alarm Systems 2011 Third Edition Central Station Alarm Association 8150 Leesburg Pike Suite 700 Vienna, VA 22182.
- [2] A.I.C.E (1999): "American Institute of Chemical Engineers". 95(7): 137.
- [3] Bernard Kleane and Russell Sander (2007): "Structural Firefighting", Strategy and Tactics: 1st edition Jones & Bartlett Learning.
- [4] Chen P. and Jiang X. (2008) Design and Implementation of Remote Monitoring System Based on GSM. Proceedings of IEEE conference on Computational Intelligence and Industrial Application, 2008. PACIIA '08. Pacific-Asia Workshop on 1:678-681.
- [5] FULLWIKIPEDIA (2014) [Online] Available from: <http://www.thefullwiki.org/Microcontroller>
- [6] Guylène Proulx 2000. Why Building Occupants Ignore Fire Alarms. Construction Technology Update No. 42 © National Research Council of Canada.
- [7] Heath, Steve 2003. Embedded systems design. EDN series for design engineers (2 ed.). Newnes. pp. 11– 12. ISBN 9780750655460.
- [8] Mehta V.K (2003): "Principles of Electronics". Published By S.Chand & Company Ltd.
- [9] Puran G., Bhausahab S., and Sagar S. (2014) Review on Gas Leak Detection Techniques. International Journal of

scientific Engineering and Technology Research. 3(15): 3204-3207.

[10] Ramya V. and Palaniappan B. (2012) Embedded system for Hazardous gas detection and alerting. International Journal of Distributed and parallel systems (IJDPA), 3:287-300

[11] Song L. and Xiong H. (2011) Advances in Computer Science, Environment, Ecoinformatics and Education”, Part II: International Conference CSEE: 331.

[12] Soundarya T., Anchitalagammai J.V, Deepa P., and Karthick Kumar S.S. (2014) C-leakage: Cylinder LPG Gas leakage Detection for home safety. IOSR Journal of Electronics and Communication (IOSR-JECE). 9(1):53- 58

[13] WIKIPEDIA (2014) [Online] Available from: <http://en.wikipedia.org/wiki/Microcontroller>

[14] WIKIPEDIA (2014) [Online] Available from: <http://en.wikipedia.org/wiki/Security>

[15] Xie Z. Tan Q. (2006) A Mini Multi-Gas Detection System Based on Infrared Principle, International Journal Mill waves. Pp. 1639-1649.

